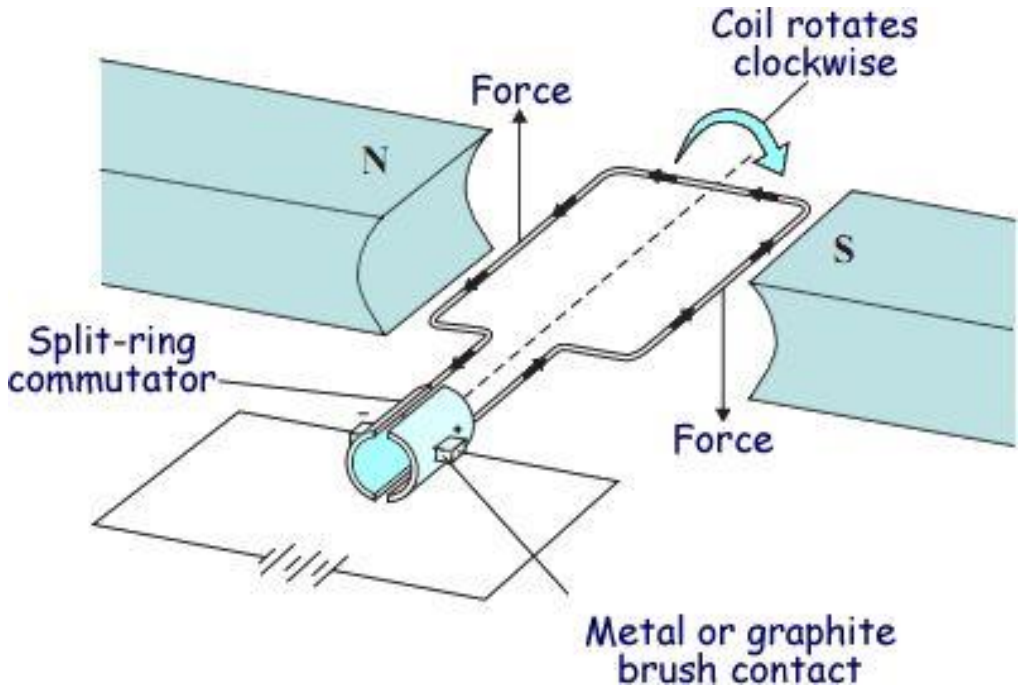
2102222 ENGINEERING ELECTROMAGNETICS



STATIC MAGNETIC FIELDS (Part 4)

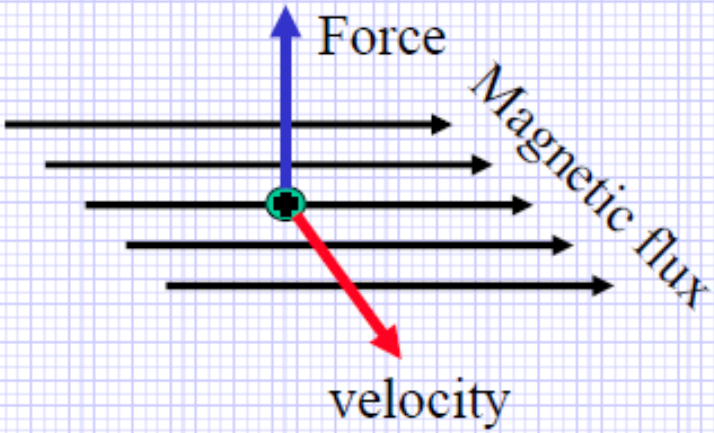
Objectives

- To understand how magnetic force exerted on current carrying conductor in magnetic field
- Be able to analyze the force acting on current carrying conductor in magnetic field.
- Be able to use Ampere's force law for calculating the magnetic force acting on current carrying conductors due to magnetic field of other current source.
- To understand how to make current loop rotate in uniform magnetic field, and be able to analyze torque.



Back to basic from physics

- We know that there is a magnetic force acting to electric charge moving across magnetic field.
- This is called "Lorentz's force"
- The equation is very simple.
- Here is the equation of magnetic force!



$$\vec{F} = q\vec{v} \times \vec{B}$$

How can we apply this basic physical law in electrical engineering ?

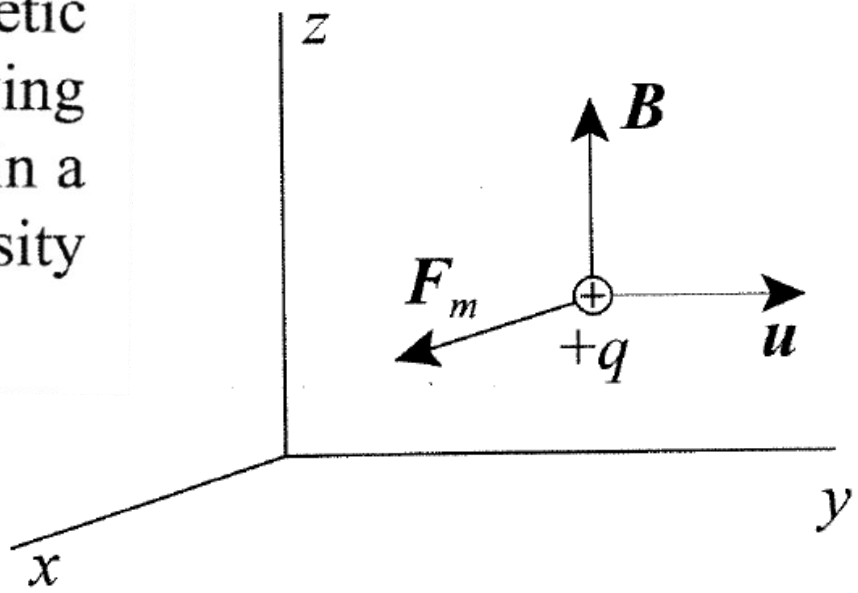
Moving Charge in E-Field & H-Field

- Given a charge moving in an electric field and a magnetic field, the total force on the charge is the superposition of electric and magnetic forces.
- This total force equation is known as the ***Lorentz force equation***.
- The total force (Lorentz force) on the point charge is given by

$$\vec{F} = \vec{F}_e + \vec{F}_m = q(\vec{E} + \vec{v} \times \vec{B})$$

Example

Determine the vector magnetic force on a point charge $+q$ moving at a uniform velocity $\boldsymbol{u} = u_o \hat{\boldsymbol{y}}$ in a uniform magnetic flux density defined by $\boldsymbol{B} = B_o \hat{\boldsymbol{z}}$.



$$\boldsymbol{F}_m = q(\boldsymbol{u} \times \boldsymbol{B})$$

$$= q[(u_o \hat{\boldsymbol{y}}) \times (B_o \hat{\boldsymbol{z}})]$$

$$= q u_o B_o \hat{\boldsymbol{x}}$$

Example

An electron with a speed of 8×10^6 m/s is projected along the positive x-direction into a medium containing a uniform magnetic flux density $\mathbf{B} = (\hat{\mathbf{x}}4 - \hat{\mathbf{z}}3)$ T. Given that $e = 1.6 \times 10^{-19}$ C and the mass of an electron is $m_e = 9.1 \times 10^{-31}$ kg, determine the initial acceleration vector of the electron (at the moment it is projected into the medium).

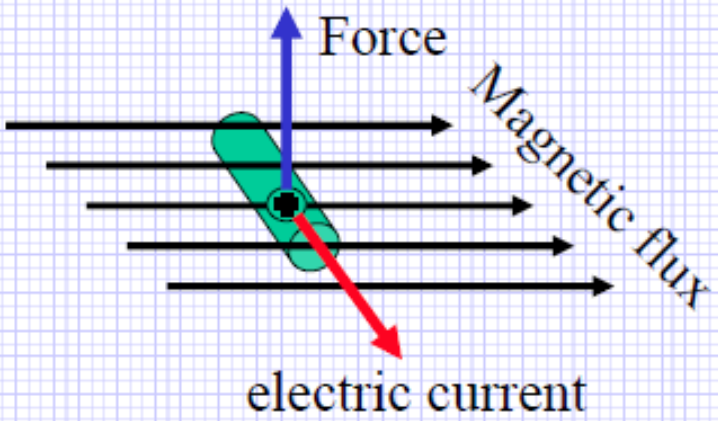
Solution: The acceleration vector of a free particle is the net force vector divided by the particle mass.

$$\mathbf{a} = \frac{\mathbf{F}_m}{m_e} = \frac{-e}{m_e}\mathbf{u} \times \mathbf{B} = \frac{-1.6 \times 10^{-19}}{9.1 \times 10^{-31}}(\hat{\mathbf{x}}8 \times 10^6) \times (\hat{\mathbf{x}}4 - \hat{\mathbf{z}}3)$$

$$= -\hat{\mathbf{y}}4.22 \times 10^{18} \quad (\text{m/s}^2).$$

Magnetic force acting to line current element

- Let us consider conductor carrying electric current.
- This is a phenomena of moving charge in conductor.
- What happen if we place current carrying conductor in magnetic field?
- The answer is very simple!
- There must be magnetic force acting to current carrying conductor.
- Are there any experiments showing this phenomena?
- Oh, yes.



Let us find the force formula

$$d\vec{F} = q\vec{v} \times \vec{B}$$

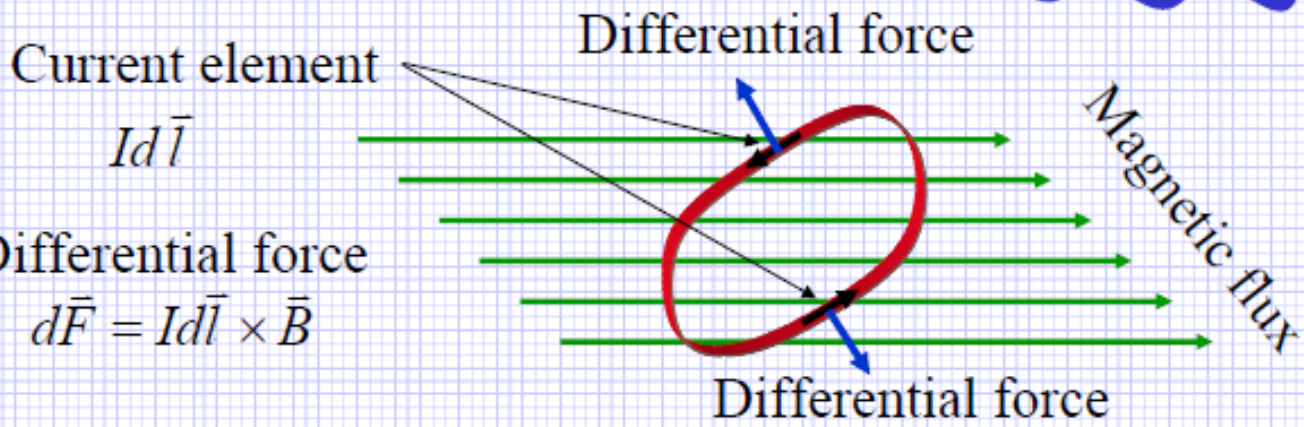
$$d\vec{F} = q\left(\frac{d\vec{l}}{\Delta t}\right) \times \vec{B}$$

$$d\vec{F} = \left(\frac{q}{\Delta t}\right) d\vec{l} \times \vec{B}$$

$$d\vec{F} = Id\vec{l} \times \vec{B}$$

$$\vec{F} = Id\vec{l} \times \vec{B}$$

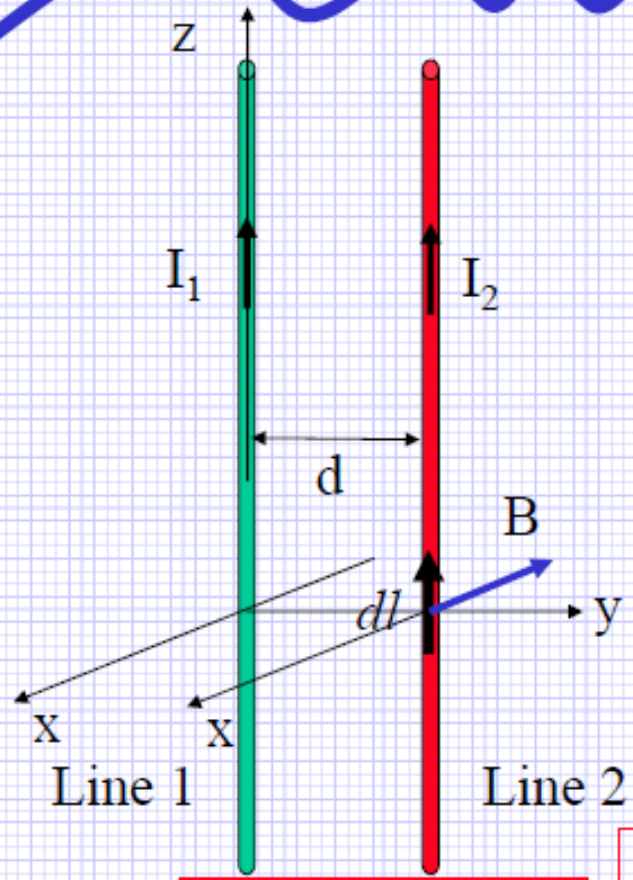
Magnetic force acting to line current loop



- The total force acting to a line current loop can be analyze as a summation of force acting to all current element.
- This is the integration of differential force acting at all point on line current loop.
- We can write the magnetic force formula acting on current loop as.

$$\bar{F}_m = \oint_C Id\bar{l} \times \bar{B}$$

Example 1 : Force between perpendicular infinite current lines



- Let us consider the problem of analyzing magnetic force per unit length exerting on each infinitely long line current.

- First, find the force acting to line 2 due to magnetic flux originating from current in line 1.

$$\vec{F}_2 = \int_0^1 I_2 d\vec{l}_2 \times \vec{B}_{21}$$

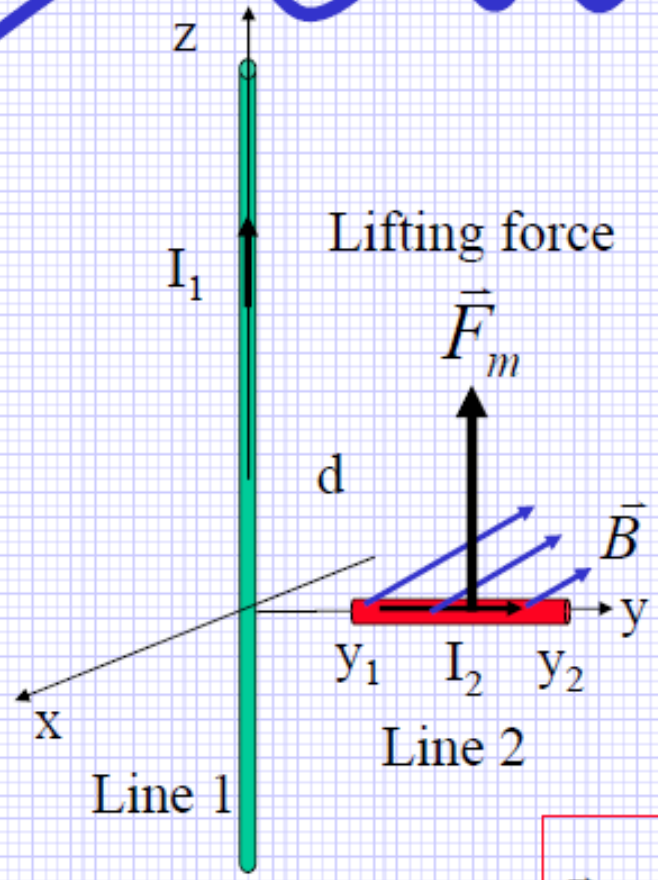
$$d\vec{l}_2 = dz\vec{a}_z, \vec{B}_{21} = -\frac{\mu_0 I_1}{2\pi d}\vec{a}_x$$

Attractive force

Answer

$$\vec{F}_2 = \int_0^1 I_2 dz\vec{a}_z \times \left(-\frac{\mu_0 I_1}{2\pi d}\vec{a}_x\right) = -\frac{\mu_0 I_1 I_2}{2\pi d}\vec{a}_y$$

Example 2 : Force between infinite line current and finite length line current



- Let us consider the problem of analyzing magnetic force exerting on finite length line current which is perpendicular to an infinite line current.
- First, find the force acting to line 2 due to magnetic flux originating from current in line 1.

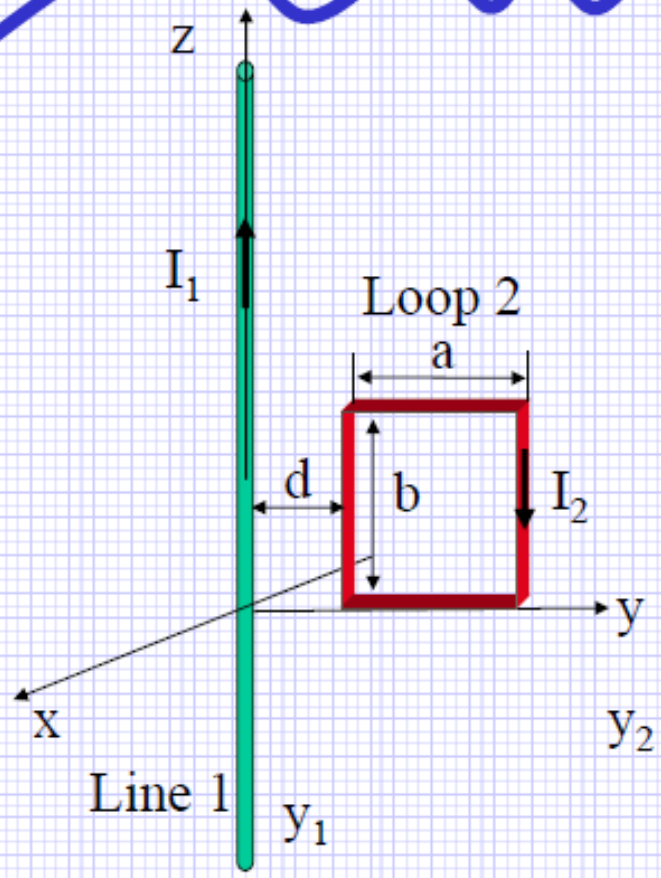
$$\vec{F}_2 = \int I_2 d\vec{l}_2 \times \vec{B}_{21}$$

$$d\vec{l}_2 = dy\bar{a}_y, \vec{B}_{21} = -\frac{\mu_0 I_1}{2\pi y}\bar{a}_x$$

$$\vec{F}_2 = \int_{y_2}^{y_1} I_2 dy\bar{a}_y \times \left(+\frac{\mu_0 I_1}{2\pi y}\bar{a}_x \right) = \frac{\mu_0 I_1 I_2}{2\pi} \ln\left(\frac{y_2}{y_1}\right)\bar{a}_z$$

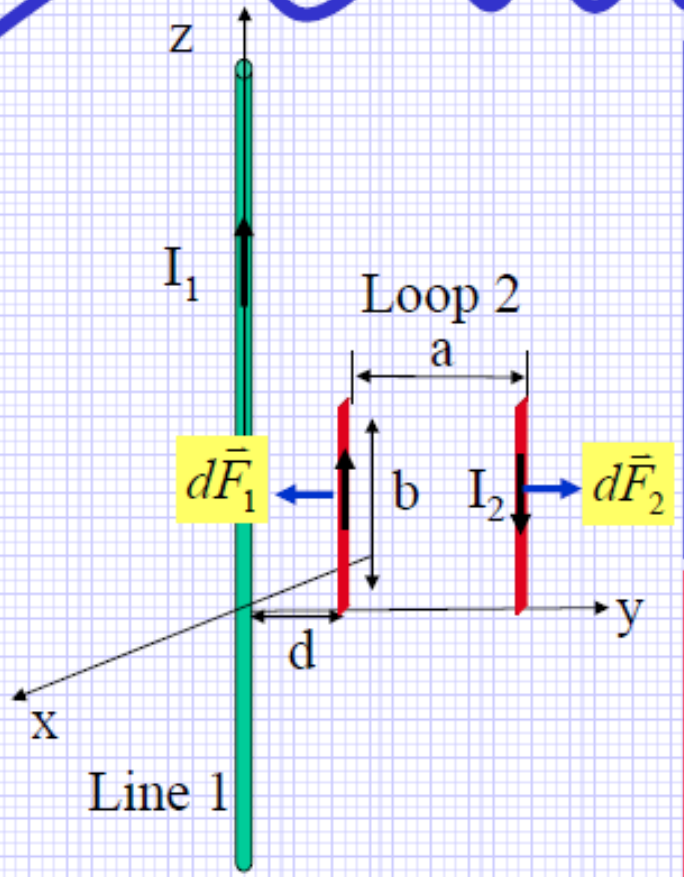
Answer

Example 3 : Force between infinite line current and rectangular loop (1/4)



- Let us consider the problem of analyzing magnetic force exerting on infinite line current and rectangular loop as shown in the figure.
- First, find the force acting to loop 2 due to magnetic flux originating from current in line 1 by dividing the loop into 4 sections.
- Find the force acting to each straight line current section and then sum all forces together.

Example 3 : Force between infinite line current and rectangular loop (2/4)



$$\vec{F}_1 = \int_0^b I_2 dz \vec{a}_z \times \vec{B}_{21}\Big|_{y=d}$$

$$\vec{B}_{21}\Big|_{y=d} = -\frac{\mu_0 I_1}{2\pi d}\vec{a}_x$$

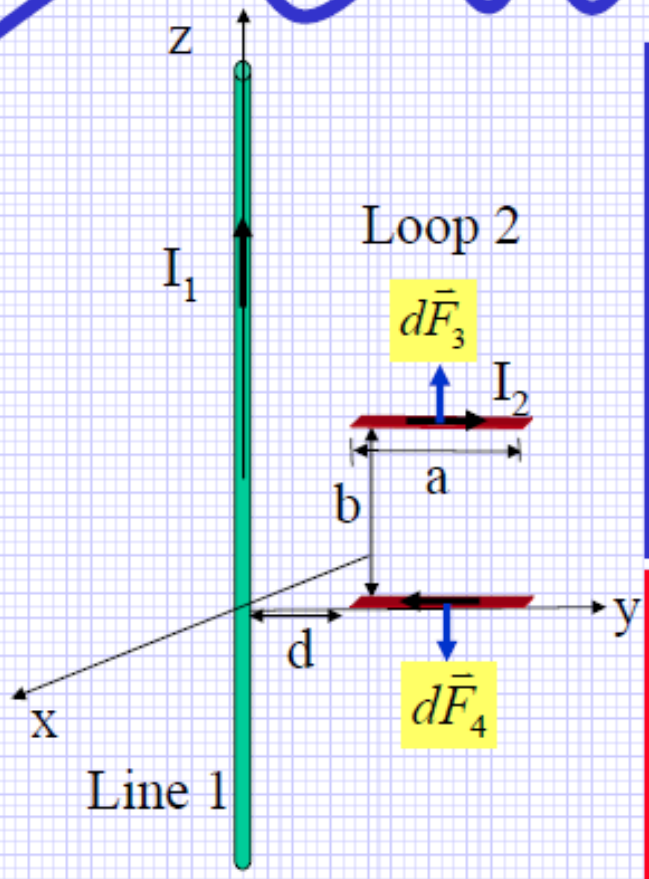
$$\vec{F}_1 = \int_0^b I_2 dz \vec{a}_z \times \left(-\frac{\mu_0 I_1}{2\pi d}\vec{a}_x\right) = -\frac{\mu_0 I_1 I_2 b}{2\pi d}\vec{a}_y$$

$$\vec{F}_2 = \int_b^0 I_2 dz \vec{a}_z \times \vec{B}_{21}\Big|_{y=d+a}$$

$$\vec{B}_{21}\Big|_{y=d+a} = -\frac{\mu_0 I_1}{2\pi(d+a)}\vec{a}_x$$

$$\vec{F}_2 = \int_b^0 I_2 dz \vec{a}_z \times \left(-\frac{\mu_0 I_1}{2\pi(d=a)}\vec{a}_x\right) = \frac{\mu_0 I_1 I_2 b}{2\pi(d+a)}\vec{a}_y$$

Example 3 : Force between infinite line current and rectangular loop (3/4)



$$\vec{F}_3 = \int_d^{d+a} I_2 dy \vec{a}_y \times \vec{B}_{21}$$

$$\vec{B}_{21} = -\frac{\mu_0 I_1}{2\pi y}\vec{a}_x$$

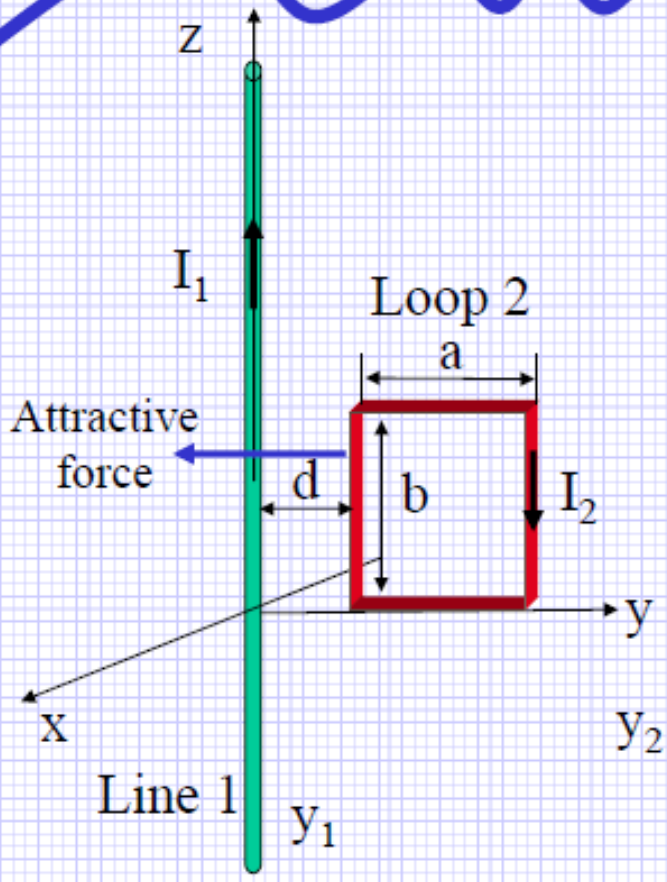
$$\vec{F}_3 = \int_d^{d+a} I_2 dy \vec{a}_y \times \left(-\frac{\mu_0 I_1}{2\pi y}\vec{a}_x\right) = \frac{\mu_0 I_1 I_2}{2\pi}\ln\left(\frac{d+a}{d}\right)\vec{a}_z$$

$$\vec{F}_4 = \int_{d+a}^{d} I_2 dy \vec{a}_y \times \vec{B}_{21}$$

$$\vec{B}_{21} = -\frac{\mu_0 I_1}{2\pi y}\vec{a}_x$$

$$\vec{F}_4 = \int_{d+a}^{d} I_2 dy \vec{a}_y \times \left(-\frac{\mu_0 I_1}{2\pi y}\vec{a}_x\right) = -\frac{\mu_0 I_1 I_2}{2\pi}\ln\left(\frac{d+a}{d}\right)\vec{a}_z$$

Example 3 : Force between infinite line current and rectangular loop (4/4)



$$\vec{F}_1 = -\frac{\mu_0 I_1 I_2 b}{2\pi d}\vec{a}_y$$

$$\vec{F}_2 = \frac{\mu_0 I_1 I_2 b}{2\pi (d+a)}\vec{a}_y$$

$$\vec{F}_3 = \frac{\mu_0 I_1 I_2}{2\pi}\ln\left(\frac{d+a}{d}\right)\vec{a}_z$$

ANS

$$\vec{F}_4 = -\frac{\mu_0 I_1 I_2}{2\pi}\ln\left(\frac{d+a}{d}\right)\vec{a}_z$$

$$\vec{F}_3 = -\vec{F}_4$$

$$\vec{F}_{total} = \vec{F}_1 + \vec{F}_2 = -\frac{\mu_0 I_1 I_2 b}{2\pi}\left(\frac{a}{d(d+a)}\right)\vec{a}_y$$

Magnetic force on a closed circuit C_1 that carries current I_1 in the presence of magnetic field caused by current I_2 in a closed circuit C_2.

$$\vec{F}_{21} = I_1 \oint_{C_1} \overrightarrow{dl_1} \times \vec{B}_{21} \xrightarrow[\vec{B}_{21} = \frac{\mu_0 I_2}{4\pi} \oint_{C_2} \frac{\overrightarrow{dl_2} \times \vec{a}_{R_{21}}}{R_{21}^2}]{\text{Biot-Savart law}}$$

$$= \frac{\mu_0}{4\pi} I_1 I_2 \oint_{C_1} \oint_{C_2} \frac{\overrightarrow{dl_1} \times \left(\overrightarrow{dl_2} \times \vec{a}_{R_{21}} \right)}{R_{21}^2}$$

$$= -\frac{\mu_0}{4\pi} I_1 I_2 \oint_{C_1} \oint_{C_2} \frac{\vec{a}_{R_{21}} \left(\overrightarrow{dl_1} \bullet \overrightarrow{dl_2} \right)}{R_{21}^2}$$

Use vector identity:

$$A \times (B \times C) = (A \cdot C)B - (A \cdot B)C$$

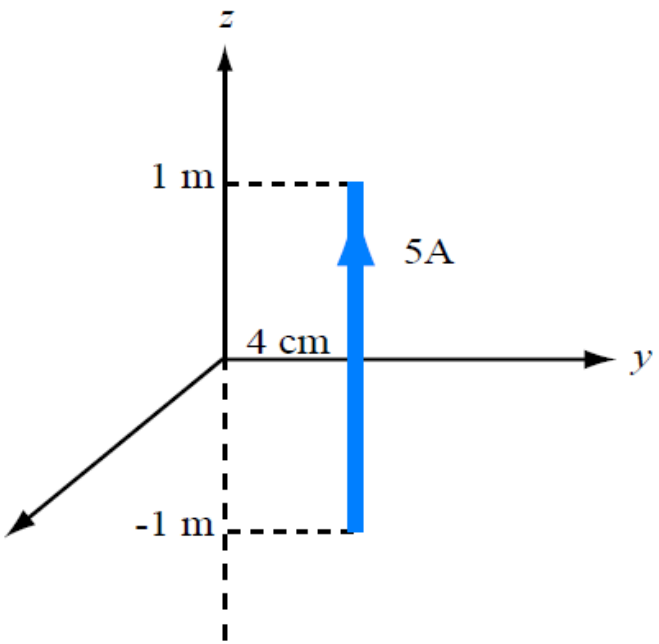
Example

In a cylindrical coordinate system, a 2-m-long straight wire carrying a current of 5 A in the positive z-direction is located at $r = 4$ cm, $\phi = \pi/2$, and $-1 \text{ m} \leq z \leq 1 \text{ m}$.

(a) If $\mathbf{B} = \hat{\mathbf{r}}\, 0.2 \cos\phi$ (T), what is the magnetic force acting on the wire?

(b) How much work is required to rotate the wire once about the z-axis in the negative ϕ-direction (while maintaining $r = 4$ cm)?

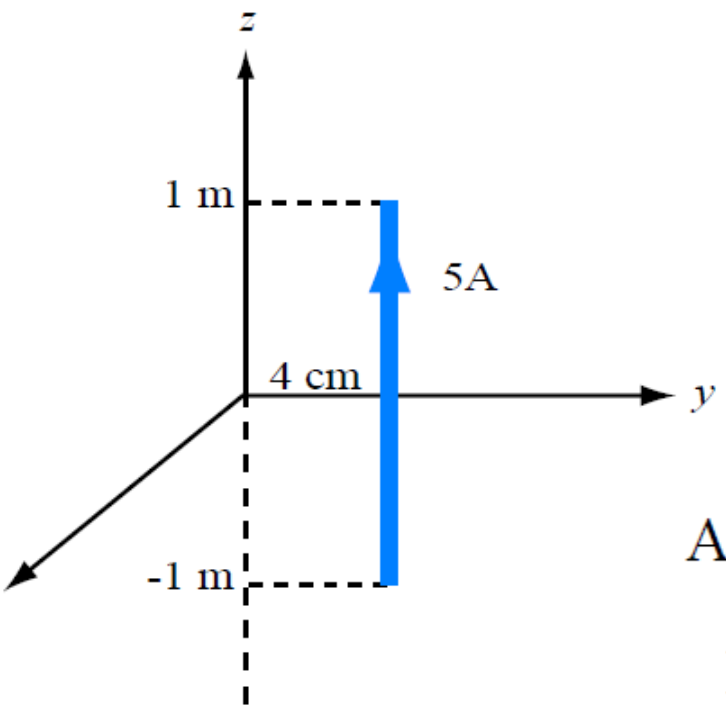
(c) At what angle ϕ is the force a maximum?



Example (cont.)

Solution:

(a)



$$\mathbf{F} = I\boldsymbol{\ell} \times \mathbf{B}$$
$$= 5\hat{\mathbf{z}}2 \times [\hat{\mathbf{r}}\,0.2\cos\phi]$$
$$= \hat{\boldsymbol{\phi}}\,2\cos\phi.$$

At $\phi = \pi/2$, $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}}$. Hence,

$$\mathbf{F} = -\hat{\mathbf{x}}\,2\cos(\pi/2) = 0.$$

Example (cont.)

(b)

$$W = \int_{\phi=0}^{2\pi} \mathbf{F} \cdot d\mathbf{l} = \int_0^{2\pi} \hat{\boldsymbol{\phi}}\,[2\cos\phi] \cdot (-\hat{\boldsymbol{\phi}})\, r\, d\phi \Big|_{r=4\text{ cm}}$$

$$= -2r \int_0^{2\pi} \cos\phi\, d\phi \Big|_{r=4\text{ cm}} = -8 \times 10^{-2}\,[\sin\phi]_0^{2\pi} = 0.$$

The force is in the $+\hat{\boldsymbol{\phi}}$-direction, which means that rotating it in the $-\hat{\boldsymbol{\phi}}$-direction would require work. However, the force varies as $\cos\phi$, which means it is positive when $-\pi/2 \le \phi \le \pi/2$ and negative over the second half of the circle. Thus, work is provided by the force between $\phi = \pi/2$ and $\phi = -\pi/2$ (when rotated in the $-\hat{\boldsymbol{\phi}}$-direction), and work is supplied for the second half of the rotation, resulting in a net work of zero.

(c) The force **F** is maximum when $\cos\phi = 1$, or $\phi = 0$.

Magnetic Torque on Current Loop

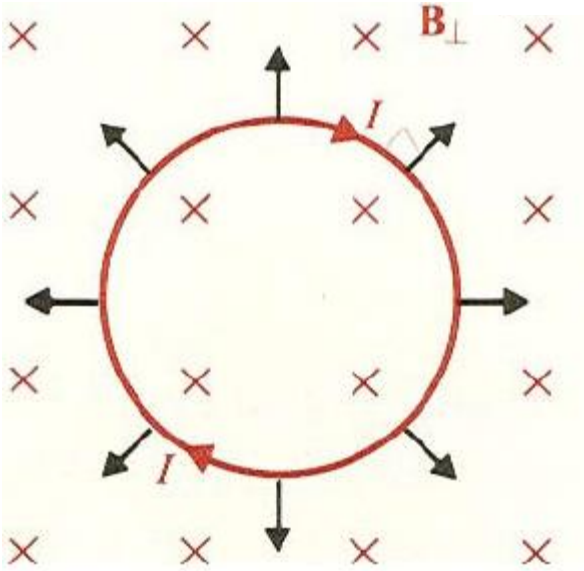
$$\vec{T} = \vec{m} \times \vec{B} \quad [\mathrm{N \cdot m}]$$

$\{\vec{B}$ must be uniform over the current loop$\}$

magnetic dipole moment: $\vec{m} = \vec{a}_n I(\text{Area})$

Ex.: circular loop w/current I in uniform $\vec{B} = \vec{B}_\perp + \vec{B}_\parallel$

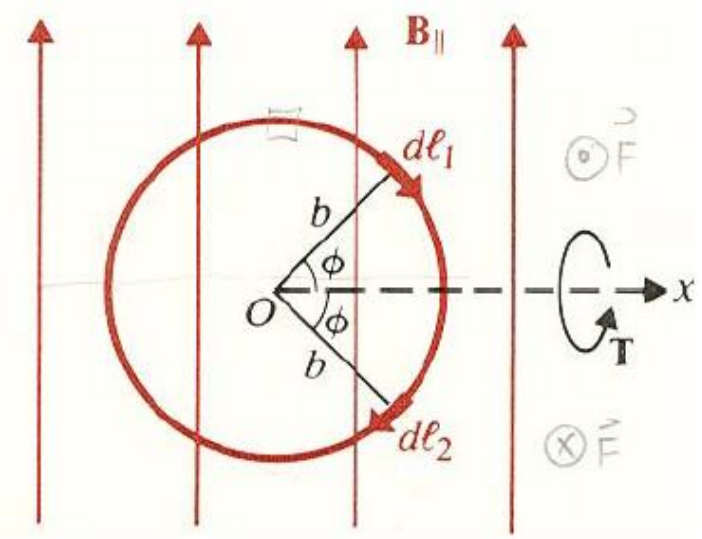
$\vec{B}_\perp$ expand the loop, but no net force to move



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Magnetic Torque on Current Loop (cont.)

$\vec{B}_{\|}$ rotate the loop around $\vec{a}_x$ to align $\vec{B}_{\text{from } I}$ with $\vec{B}_{\|}$



$d\vec{T}$ produced by $d\vec{F}_1$ (out) & $d\vec{F}_2$ (in):

$$d\vec{T} = \vec{a}_x \left(dF_1 + dF_2\right)\left(b\sin\phi\right)$$

$$\vec{a}_x \left(Idl\, B_{\|}\sin\phi\right) 2\left(b\sin\phi\right) \xrightarrow{dl = b\,d\phi} \vec{a}_x\, 2Ib^2 B_{\|}\sin^2\phi\, d\phi$$

$$\therefore \quad \vec{T} = \int d\vec{T} = \vec{a}_x\, 2Ib^2 B_{\|}\int_0^{\pi}\sin^2\phi d\phi \xrightarrow{\int \sin^2 ax dx = \frac{x}{2} - \frac{\sin 2ax}{4a}}$$

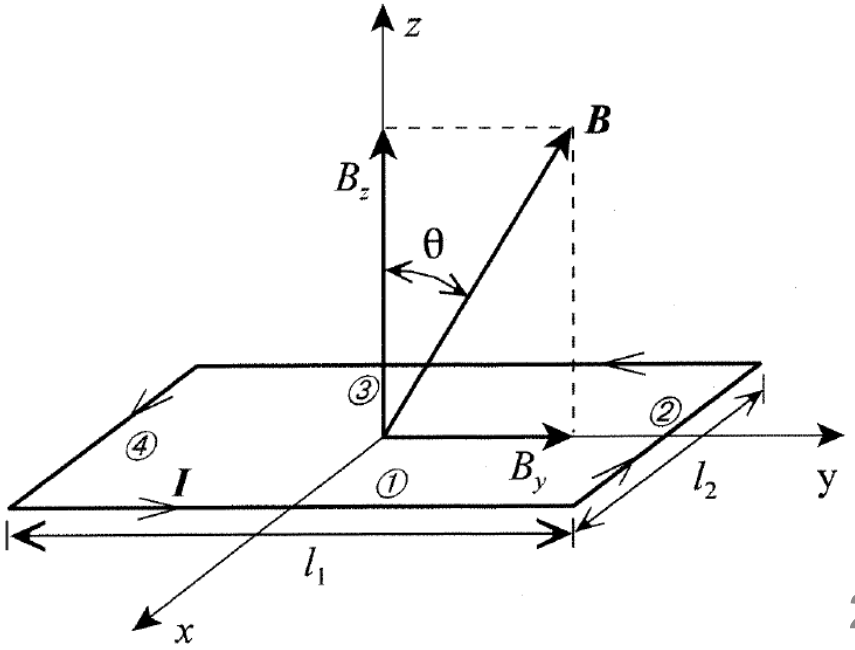
$$= \vec{a}_x\, 2Ib^2 B_{\|}\left[\frac{\pi}{2}\right] = \vec{a}_x I(\pi b^2) B_{\|} = \vec{m}\times\vec{B}$$

Magnetic Torque on Current Loop (cont.)

Given the change in current directions around a closed current loop, the magnetic forces on different portions of the loop vary in direction. Using the Lorentz force equation, we can show that the net force on a simple circular or rectangular loop is a torque which forces the loop to align its *magnetic moment* with the applied magnetic field.

Consider the rectangular current loop shown below. The loop lies in the *x*-*y* plane and carries a DC current *I*. The loop lies in a uniform magnetic flux density $\boldsymbol{B}$ given by

$$\boldsymbol{B} = B_y \hat{\boldsymbol{y}} + B_z \hat{\boldsymbol{z}}$$



Magnetic Torque on Current Loop (cont.)

The loop consists of four distinct vector current segments.

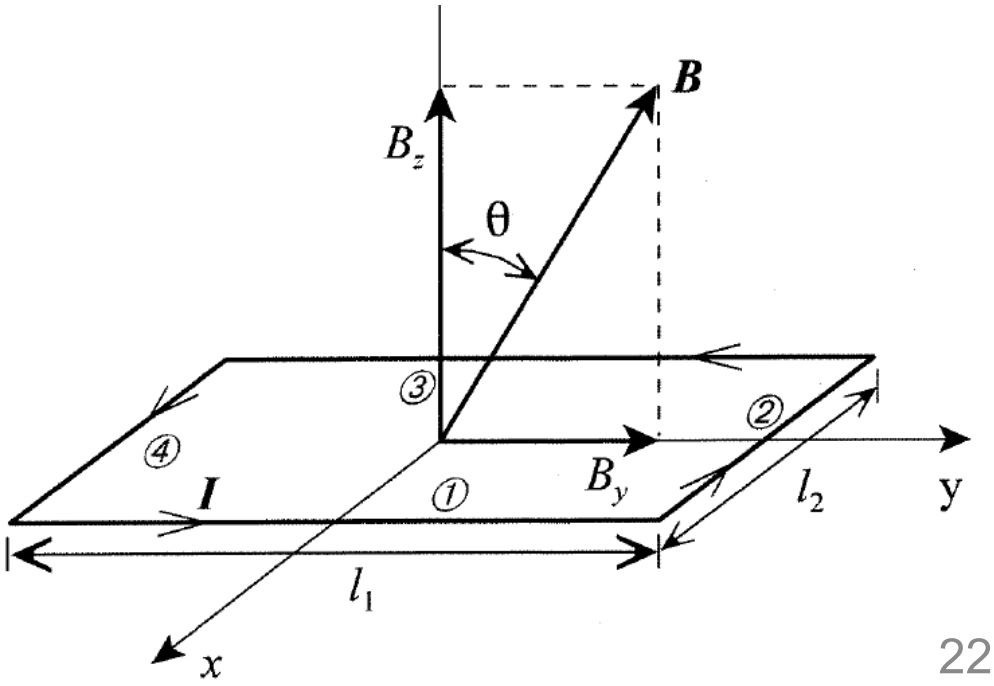
$$\boldsymbol{I}_1 = I\hat{\boldsymbol{y}} \qquad \boldsymbol{I}_2 = -I\hat{\boldsymbol{x}} \qquad \boldsymbol{I}_3 = -I\hat{\boldsymbol{y}} \qquad \boldsymbol{I}_4 = I\hat{\boldsymbol{x}}$$

Given a uniform flux density and a DC current along straight current segments, the magnetic force on each conductor segment can be simplified to the following equation.

$$\boldsymbol{F}_m = \int_L (\boldsymbol{I}\times\boldsymbol{B})\,dl'$$

$$= (\boldsymbol{I}\times\boldsymbol{B})\int_L dl'$$

$$= (\boldsymbol{I}\times\boldsymbol{B})L$$



Magnetic Torque on Current Loop (cont.)

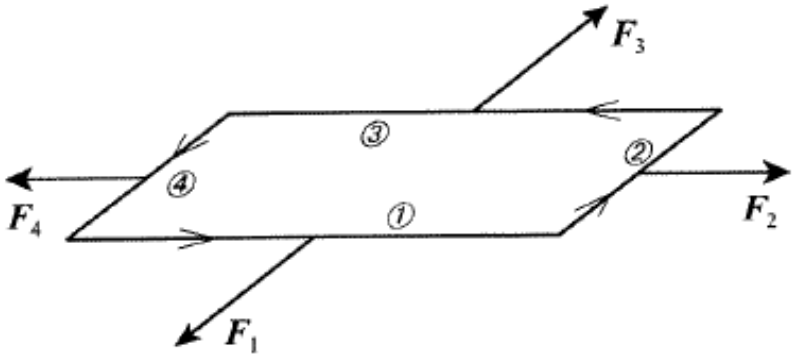
Forces due to B_z (net force = 0)

$$\boldsymbol{F}_1 = (I\hat{\boldsymbol{y}} \times B_z\hat{\boldsymbol{z}})\,l_1 = IB_z l_1\hat{\boldsymbol{x}}$$

$$\boldsymbol{F}_2 = (-I\hat{\boldsymbol{x}} \times B_z\hat{\boldsymbol{z}})\,l_2 = IB_z l_2\hat{\boldsymbol{y}}$$

$$\boldsymbol{F}_3 = (-I\hat{\boldsymbol{y}} \times B_z\hat{\boldsymbol{z}})\,l_1 = -IB_z l_1\hat{\boldsymbol{x}}$$

$$\boldsymbol{F}_4 = (I\hat{\boldsymbol{x}} \times B_z\hat{\boldsymbol{z}})\,l_2 = -IB_z l_2\hat{\boldsymbol{y}}$$



Magnetic Torque on Current Loop (cont.)

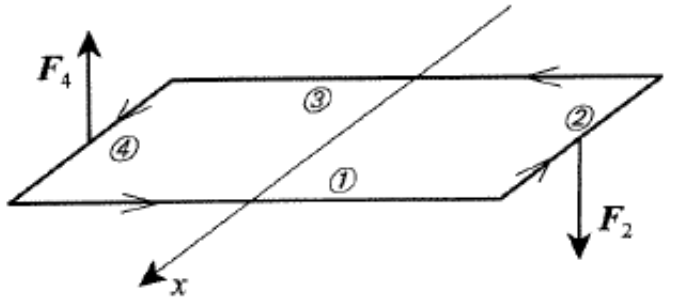
Forces due to B_y (net force = vector torque $\boldsymbol{T}$)

$$\boldsymbol{F}_1 = (I\hat{\boldsymbol{y}} \times B_y \hat{\boldsymbol{y}}) l_1 = \boldsymbol{0}$$

$$\boldsymbol{F}_2 = (-I\hat{\boldsymbol{x}} \times B_y \hat{\boldsymbol{y}}) l_2 = -IB_y l_2 \hat{\boldsymbol{z}}$$

$$\boldsymbol{F}_3 = (-I\hat{\boldsymbol{y}} \times B_y \hat{\boldsymbol{y}}) l_1 = \boldsymbol{0}$$

$$\boldsymbol{F}_4 = (I\hat{\boldsymbol{x}} \times B_y \hat{\boldsymbol{y}}) l_2 = IB_y l_2 \hat{\boldsymbol{z}}$$



Magnetic Torque on Current Loop (cont.)

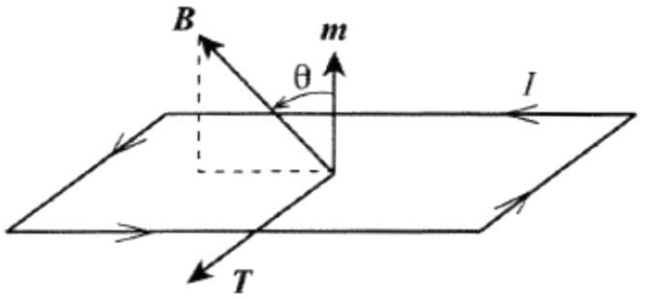
The vector torque on the loop is defined in terms of the force magnitude ($IB_y l_2$), the torque moment arm distance ($l_1/2$), and the torque direction (defined by the right hand rule):

$$\boldsymbol{T} = 2(IB_y l_2)\left(\frac{l_1}{2}\right)(-\hat{\boldsymbol{x}}) = -IB_y l_1 l_2 \hat{\boldsymbol{x}} = -IAB_y \hat{\boldsymbol{x}}$$

where $A = l_1 l_2$ is the loop area. The vector torque can be written in a general way in terms of the *vector magnetic moment* ($\boldsymbol{m}$) of the loop.

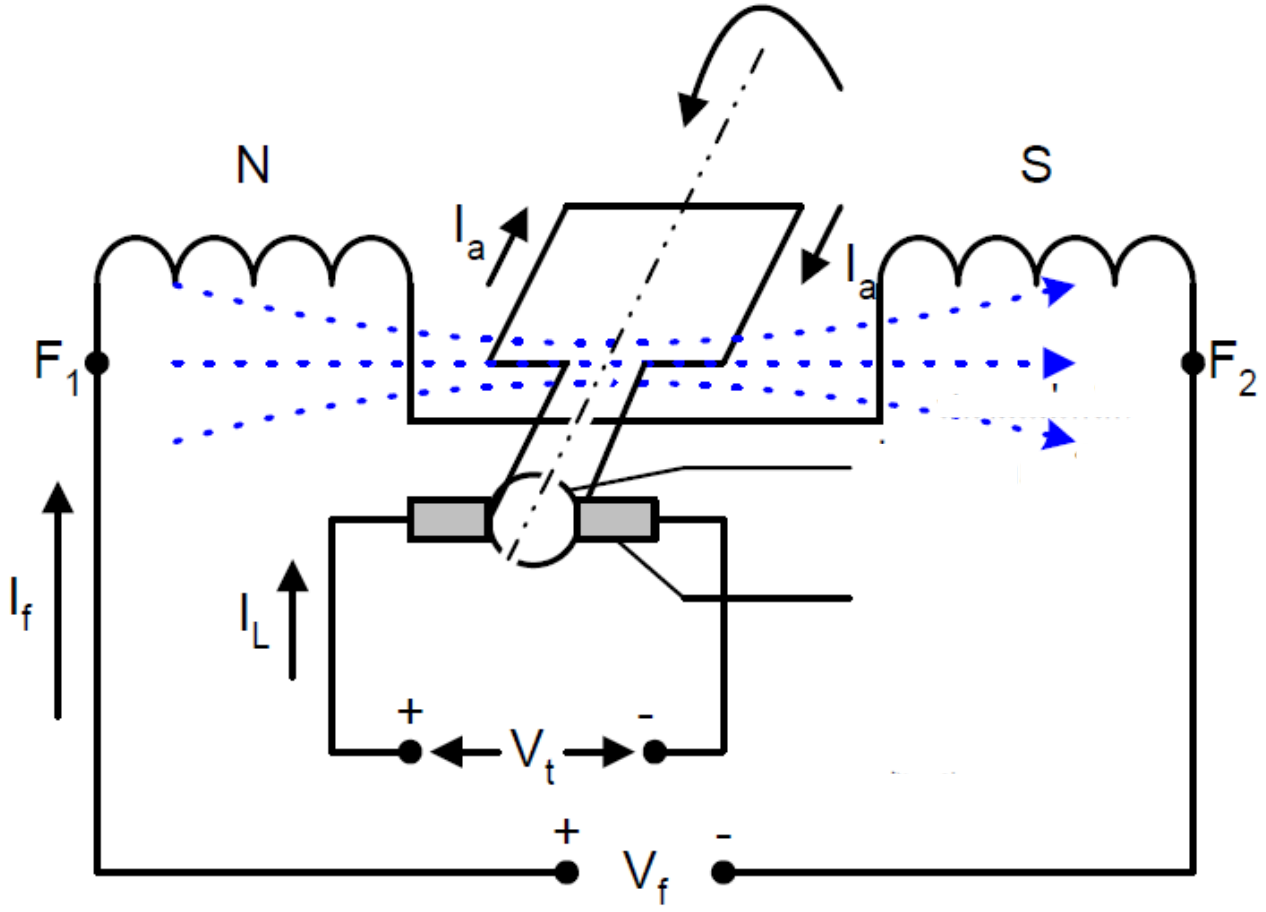
$$\boldsymbol{m} = IA\hat{\boldsymbol{n}} \qquad \text{(vector magnetic moment)}$$

where $\hat{\boldsymbol{n}}$ is the unit normal to the loop (defined by the right hand rule as applied to the current direction). The vector torque in terms of the magnetic moment is



$$\boldsymbol{T} = \boldsymbol{m} \times \boldsymbol{B} = mB\sin\theta\,\hat{\boldsymbol{a}}_T = IAB\sin\theta\,\hat{\boldsymbol{a}}_T$$

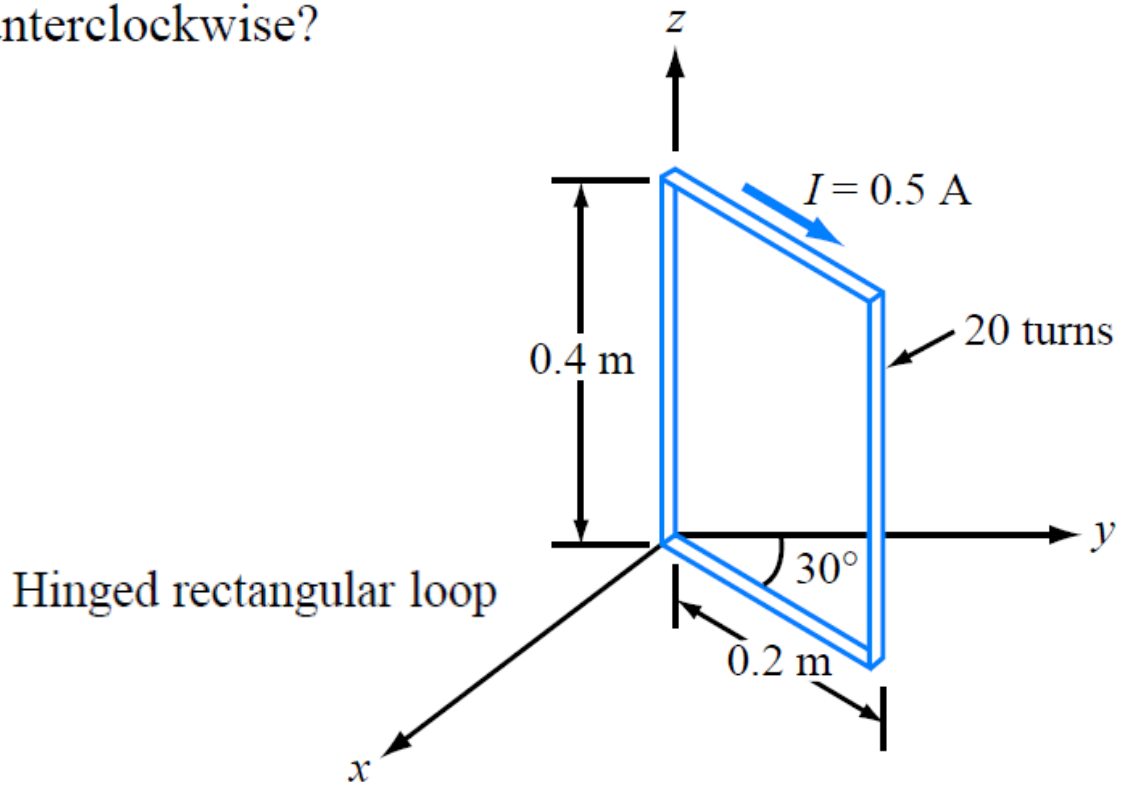
Example: DC Motor



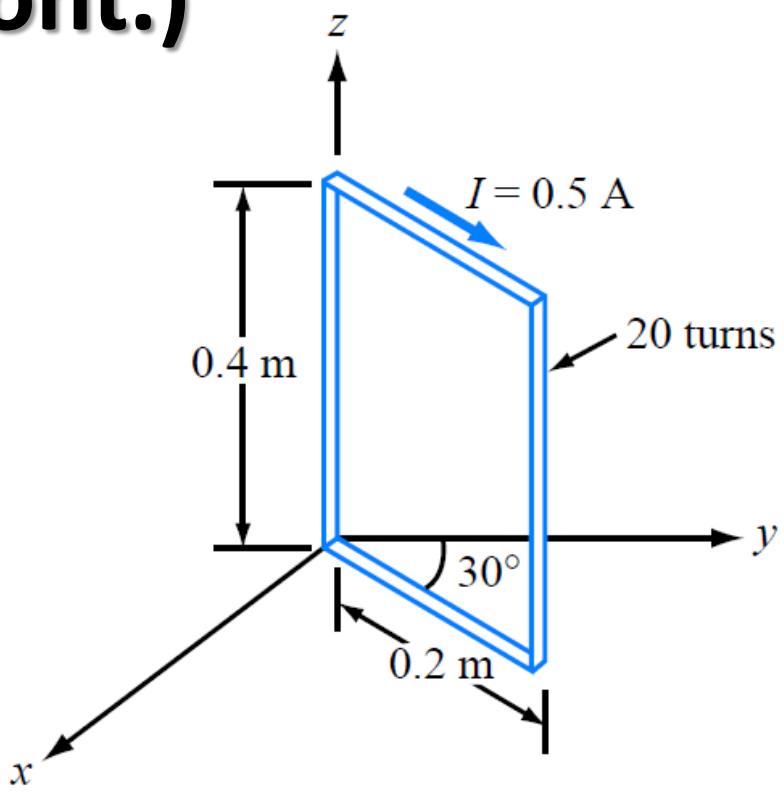
$$T = I_a B_f \sin\theta \propto I_a I_f$$

Example

The rectangular loop shown in the figure below consists of 20 closely wrapped turns and is hinged along the z-axis. The plane of the loop makes an angle of 30° with the y-axis, and the current in the windings is 0.5 A. What is the magnitude of the torque exerted on the loop in the presence of a uniform field $\mathbf{B} = \hat{\mathbf{y}}2.4$ T? When viewed from above, is the expected direction of rotation clockwise or counterclockwise?



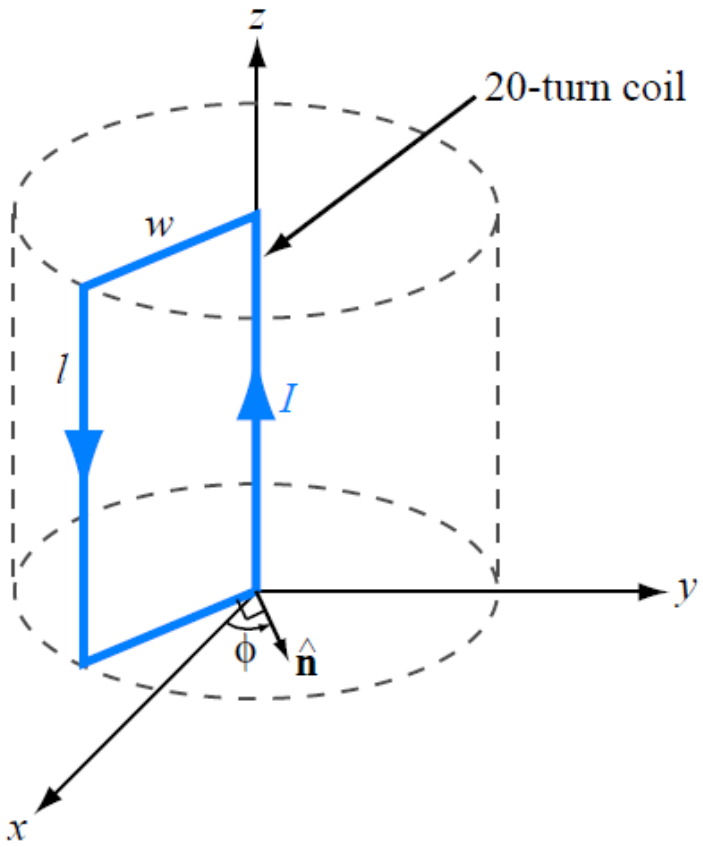
Example (cont.)



Solution: The magnetic torque on a loop is given by $\mathbf{T} = \mathbf{m} \times \mathbf{B}$, where $\mathbf{m} = \hat{\mathbf{n}}NIA$. For this problem, it is given that $I = 0.5$ A, $N = 20$ turns, and $A = 0.2\ \text{m} \times 0.4\ \text{m} = 0.08\ \text{m}^2$. From the figure, $\hat{\mathbf{n}} = -\hat{\mathbf{x}}\cos 30° + \hat{\mathbf{y}}\sin 30°$. Therefore, $\mathbf{m} = \hat{\mathbf{n}}0.8\ (\text{A}\cdot\text{m}^2)$ and $\mathbf{T} = \hat{\mathbf{n}}0.8\ (\text{A}\cdot\text{m}^2) \times \hat{\mathbf{y}}2.4\ \text{T} = -\hat{\mathbf{z}}1.66\ (\text{N}\cdot\text{m})$. As the torque is negative, the direction of rotation is clockwise, looking from above.

Example

- A 20-turn rectangular coil with side $l = 20$ cm and $w = 10$ cm is placed in the y–z plane as shown in the figure.



(a) If the coil, which carries a current $I = 10$ A, is in the presence of a magnetic flux density

$$\mathbf{B} = 2 \times 10^{-2}(\hat{\mathbf{x}} + \hat{\mathbf{y}}2) \quad \text{(T)},$$

determine the torque acting on the coil.

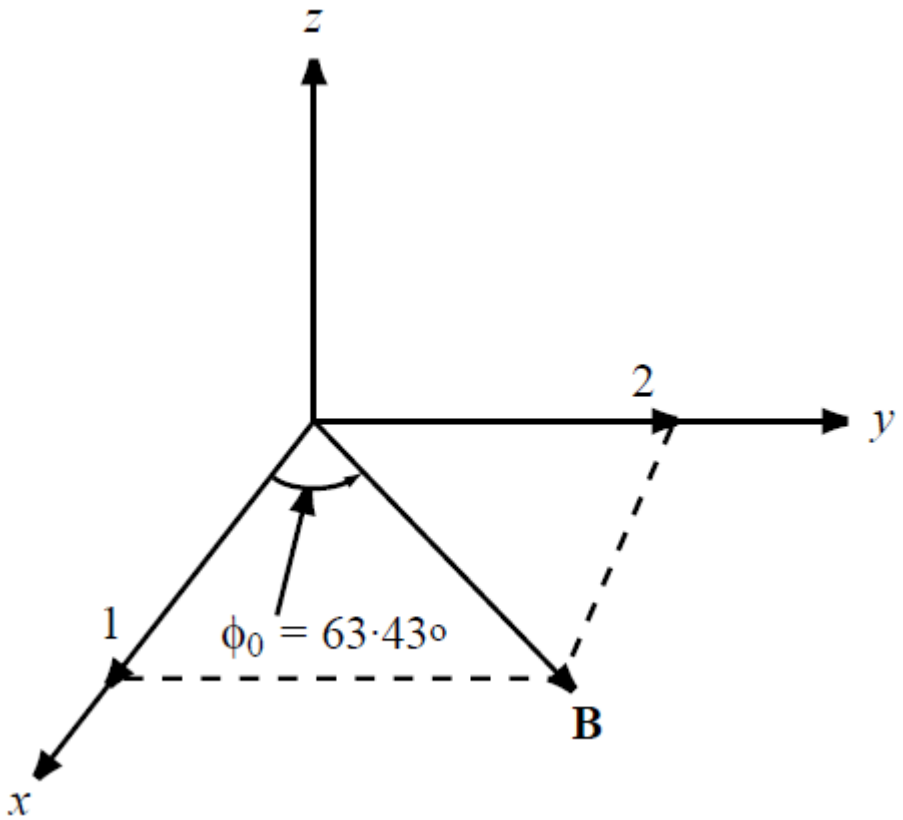
(b) At what angle ϕ is the torque zero?

(c) At what angle ϕ is the torque maximum? Determine its value.

Example (cont.)

Solution:

(a) The magnetic field is in direction $(\hat{\mathbf{x}} + \hat{\mathbf{y}}2)$, which makes an angle $\phi_0 = \tan^{-1}\frac{2}{1} = 63.43°$.



Example (cont.)

The magnetic moment of the loop is

$$\mathbf{m} = \hat{\mathbf{n}} NIA = \hat{\mathbf{n}}\, 20 \times 10 \times (30 \times 10) \times 10^{-4} = \hat{\mathbf{n}}\, 6 \quad (\mathrm{A \cdot m^2}),$$

where $\hat{\mathbf{n}}$ is the surface normal in accordance with the right-hand rule. When the loop is in the negative-y of the y–z plane, $\hat{\mathbf{n}}$ is equal to $\hat{\mathbf{x}}$, but when the plane of the loop is moved to an angle ϕ, $\hat{\mathbf{n}}$ becomes

$$\begin{aligned}
\hat{\mathbf{n}} &= \hat{\mathbf{x}} \cos\phi + \hat{\mathbf{y}} \sin\phi, \\
\mathbf{T} &= \mathbf{m} \times \mathbf{B} = \hat{\mathbf{n}}\, 6 \times 2 \times 10^{-2} (\hat{\mathbf{x}} + \hat{\mathbf{y}} 2) \\
&= (\hat{\mathbf{x}} \cos\phi + \hat{\mathbf{y}} \sin\phi)\, 6 \times 2 \times 10^{-2} (\hat{\mathbf{x}} + \hat{\mathbf{y}} 2) \\
&= \hat{\mathbf{z}}\, 0.12 [2\cos\phi - \sin\phi] \quad (\mathrm{N \cdot m}).
\end{aligned}$$

Example (cont.)

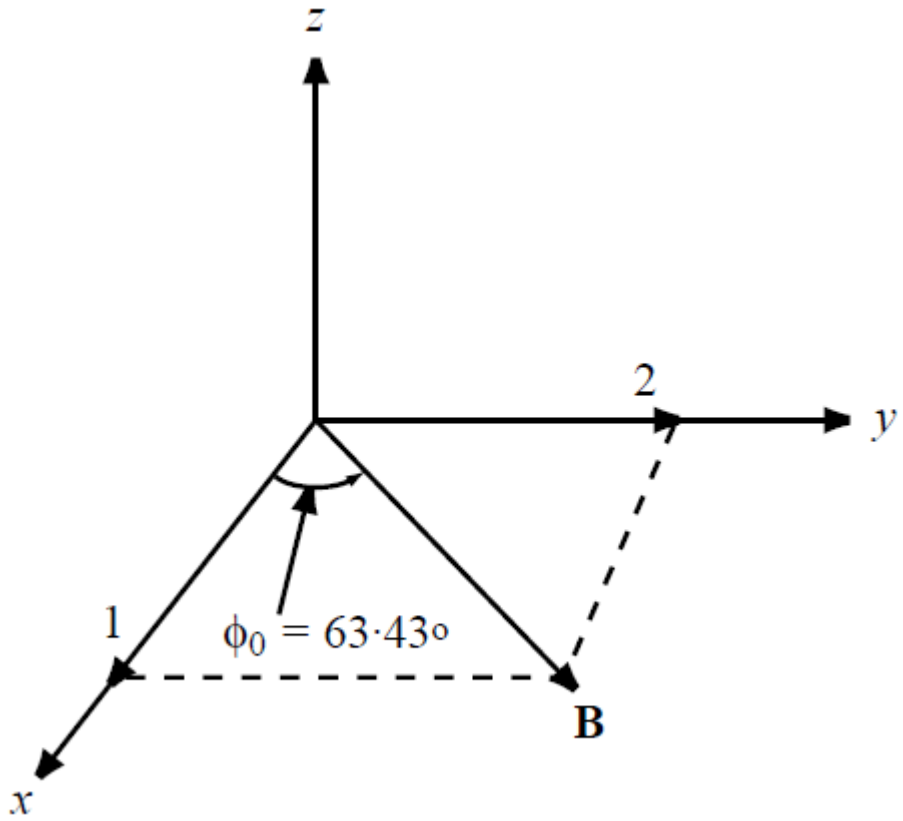
(b) The torque is zero when

$$2\cos\phi - \sin\phi = 0,$$

or

$$\tan\phi = 2,$$

$$\phi = 63.43^\circ \text{ or } -116.57^\circ.$$



Thus, when $\hat{\mathbf{n}}$ is parallel to $\mathbf{B}$, $\mathbf{T} = 0$.

Example (cont.)

(c) The torque is a maximum when $\hat{\mathbf{n}}$ is perpendicular to $\mathbf{B}$, which occurs at

$$\phi = 63.43 \pm 90^\circ = -26.57^\circ \text{ or } +153.43^\circ.$$

Mathematically, we can obtain the same result by taking the derivative of $\mathbf{T}$ and equating it to zero to find the values of ϕ at which $|\mathbf{T}|$ is a maximum. Thus,

$$\frac{\partial T}{\partial \phi} = \frac{\partial}{\partial \phi}(0.12(2\cos\phi - \sin\phi)) = 0$$

or

$$-2\sin\phi + \cos\phi = 0,$$

which gives $\tan\phi = -\frac{1}{2}$, or

$$\phi = -26.57^\circ \text{ or } 153.43^\circ,$$

at which $\mathbf{T} = \hat{\mathbf{z}}\,0.27$ (N·m).